Predicting the Clear-sky and Clearness Indices for GHI Estimation under Two Algerian Climate

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***Abstract*— Solar radiation variability represents a major challenge to the optimal utilization of solar energy. This study focuses on predicting the clear sky index (K_t) and the clearness index (K_d) in two climatically different regions of Algeria: In Salah (Warm desert climate) and Souk Ahras in Oued el Kebrit (cold semi-arid climate steppe cold), according to the Koppen-Geiger climate classification. These will then be used to estimate horizontal solar radiation (GHI) using time series models. The National Solar Radiation Database (NSRDB) was used as the database for the period January 1 to December 31, 2022, with an hourly time resolution. The results indicate that both indices provide accurate and reliable forecasts in regions with stable climatic conditions, such as In Salah. However, the prediction accuracy decreases in regions with high climatic variability, such as Souk Ahras. Furthermore, the K_t index demonstrated superior performance compared to the K_d index. The main contribution of this research lies in first forecasting the indices K_d and K_t, and then using them subsequently to estimate GHI, Unlike previous studies that directly predict GHI using time series models. This approach improves data stationarity and significantly contributes to enhancing forecasting accuracy.**

***Keywords*— global horizontal irradiance, clear sky index, clearness index, NSRDB, statistical models, time series**

I. Introduction

With the growing global demand for electricity, reliance on traditional sources is no longer sufficient to meet these needs. Therefore, the role of renewable energy has emerged as one of the best option, led by solar energy, which is environmentally friendly, low-cost, and abundant. However, despite the importance of photovoltaic solar power plants, their intermittent production can lead to fluctuations in the supply [1].

Algeria has great potential for solar energy development, thanks to its vast area and abundant solar radiation. However, the diversity of climatic and geographic conditions across its various regions requires the use of accurate forecasting models [2] .

Several studies have predicted the Global Horizontal Irradiance (GHI), which is the key parameter for assessing photovoltaic energy production, using various models, including the ARIMA time series model [3].

This study aims to predict the K_t and K_d indices at two locations in Algeria with distinct climates: In Salah with a BWh climate and Oued el Kebrit (Souk Ahras) with BSk climate. That can be used later in estimating the GHI and characterize solar radiation variability in order to improve solar PV plants production in Algeria.

This study was based on data from the National Solar Radiation Data Base (NSRDB), one of the most widely used sources of satellite data from around the world [4].

The data covers the period from January 1, 2022, to December 31, 2022. With a time resolution of 1 hour.

K_t is defined as the ratio of GHI to the clear-sky GHI, and K_d represents the ratio of GHI to the extraterrestrial irradiance, These Two indices allow for an accurate estimation of the GHI and improved solar energy utilization [5].

$$K_t = GHI / Clear\ sky\ GHI \quad (1)$$

$$K_d = GHI / G_0 \quad (2)$$

The extraterrestrial radiation $G_0(t)$ is given as [6] :

$$G_0(t) = Esc * sin(h) * (1 + 0.03344 \cos(0.98N - 2.8)) \quad (3)$$

Where:

Esc is the solar constant =1367 w/m^2.

h is the solar altitude angle (°).

N is the day number of the year (1 to 365).

This research focuses on using K_d and K_t indices to forecast hourly GHI using an ARIMA model at two climatically distinct sites. This approach contributes to enhancing data stationary and improving forecasting accuracy, compared to previous studies that applied ARIMA directly to forecast GHI [7] or attempted to improve its performance using Machine learning or hybrids models [8], which function as a "black box," making it difficult to interpret how they work or understand the reasons behind their results

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Our approach demonstrates competitive performance using a simpler model in regions with stable climatic conditions.

From this perspective, this research aims to make a scientific contribution by relying on a simple and interpretable model to forecast GHI in two climatically different regions.

II. ARIMA MODEL METHODOLOGY

The Autoregressive Integrated Moving Average (ARIMA) model is a popular and effective model for time series analysis. It relies on three main components [9] :

- AR (Auto Regression): shows the effect of previous past values on current values.
- MA (Moving Average): uses past forecast errors to improve current forecasts,
- I (Differenced): Stabilizes a time series by removing seasonal trends and fluctuations.

The ARIMA model consists of three components [10]:

- p: represents the number of past values used to predict the current value.
- d: represents the number of times the series is differenced to become stationary.
- q: represents the number of past prediction errors used to improve the current values.

The Box-Jenkins methodology represents an iterative framework for ARIMA modeling, which involves three steps

- Identification
- Estimation
- Verification

During the first step, data were analyzed to examine seasonal trends and anomalies, and differentiation was applied to make the series stationary. Partial Autocorrelation Function (PACF) and Autocorrelation Function (ACF) plots were used to determine parameters p and q. In the second step, once the parameters were chosen, the coefficients were estimated using statistical methods. The final step involved analyzing the model residuals to ensure consistency with the white noise. Performance was evaluated using indices such as Normalized Root Mean Squared Error (NRMSE), Coefficient of determination (R^2) [11].

III. DATA SOURCE

In this study, data from the National Solar Radiation Database (NSRDB), created by the National Renewable Energy Laboratory (NREL), were used. The NSRDB is the most frequently used public database of solar irradiation data from satellites around the world, as it provides broadband irradiance estimates, including GHI, DHI and DNI ,as well as other auxiliary factors, such as solar zenith angle , cloud type and fill flag [12].

IV. DATA PRE-PROCESSING

When forecasting global solar radiation, it is common practice to filter the data in order to exclude night-time hours and enable objective comparison between the various predictors analyzed.

This choice is based on the fact that at these times, it is obvious that there is no significant solar radiation capable of producing electricity (i.e. very low potential during the night). It should also be noted that at sunrise and sunset, forecasting is particularly difficult, especially in mountainous areas, due to geographical obstacles [13].

We chose to apply a filtering criterion based on the solar zenith angle; sunshine data with a solar zenith angle exceeding 85° were eliminated.

V. RESULTS

a) In Salah

The results in Table 1 and 2 of the GHI estimates for the in Salah region using K_d index showed good performance, as the R^2 was high and the NRMSE was low, indicating that the model used was able to capture most of the changes in the GHI. This indicates that the climate In Salah is stable. Using K_t improves the results further compared to K_d, which gives greater accuracy for GHI prediction

b) Souk Ahras

In Souk Ahras, using K_d was less accurate, as R^2 was very low and NRMSE was high, indicating significant cloud cover. A slight improvement was observed when using K_t for the same region, however, the overall performance remained weak.

TABLE I. ESTIMATION OF GHI USING CLEARNESS INDEX AT IN SALAH AND SOUK AHRAS WITH PERFORMANCE EVALUATION VIA R2 AND NRMSE FOR TRAINING AND TESTING SETS

Site	Train R^2	Train NRMSE (%)	Test R^2	Test NRMSE (%)
IN SALAH	**0.96**	**10.84**	**0.90**	**18.48**
SOUK AHRAS	0.89	22.11	0.02	48.90

TABLE II. ESTIMATION OF GHI USING CLEAR SKY INDEX AT IN SALAH AND SOUK AHRAS WITH PERFORMANCE EVALUATION VIA R2 AND NRMSE FOR TRAINING AND TESTING SETS

Site	Train R^2	Train NRMSE (%)	Test R^2	Test NRMSE (%)
IN SALAH	**0.97**	**7.96**	**0.93**	**15.90**
SOUK AHRAS	0.90	21.05	0.07	47.48

Note: the last 6 hours of the data were used as the test set, while the remaining data were used for training. The best results are highlighted in bold.

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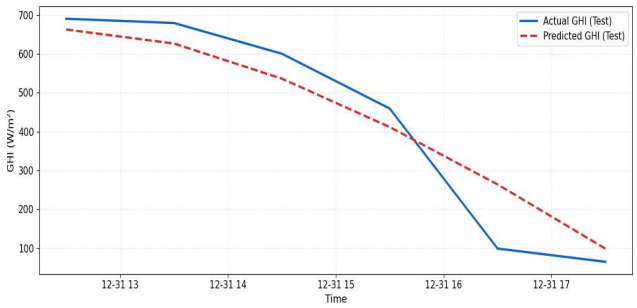


Fig. 1. Predicting GHI using the clearness index on the In Salah site

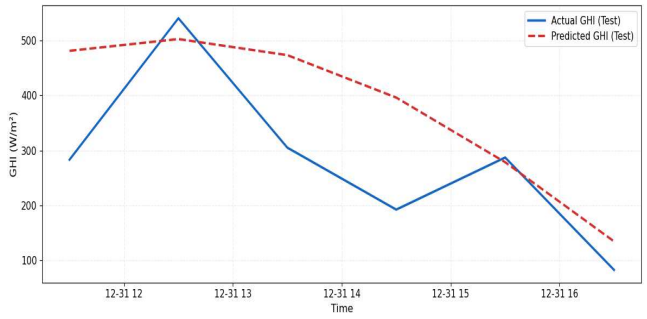


Fig. 2. Predicting GHI using the clearness index on the Souk Ahras site

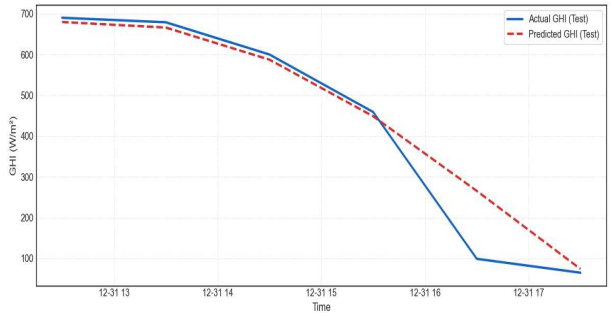


Fig. 3. Predicting GHI using the clear sky index on the In Salah site

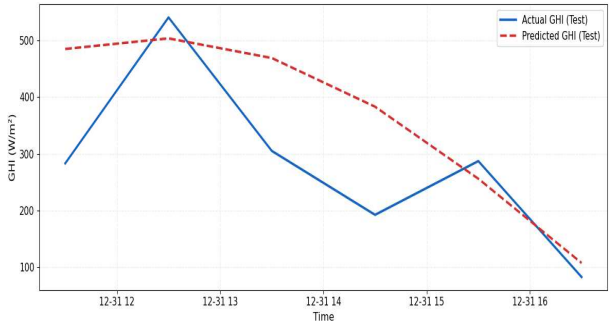


Fig. 4. Predicting GHI using the clear sky index on the Souk Ahras site

Fig. 1 shows that the K_d index achieved good results at the In Salah site, where the climatic conditions were stable. Fig. 2, for the Souk Ahras site, is less accurate and shows a clear contrast.

Fig. 3 shows a comparison between the GHI extracted from the NSRDB database and the GHI predicted using the clear sky index at In Salah. We observed a significant convergence between the two curves, which reflects the efficiency of K_t in reconstructing the GHI. In contrast, Fig. 4 shows a significant difference between the two curves. This is due to the climatic fluctuations of the Souk Ahras site. Consequently, the performance of the K_t index in Souk Ahras is less accurate compared to regions with a stable climate, such as In Salah.

CONCLUSION

This work focuses on applying the K_d and K_t indices to predict GHI using hourly data from NSRDB, covering the period from January 1, 2022, to December 31, 2022, at In Salah and Souk Ahras. These two sites were selected to represent contrasting climate within Algeria.

The ARIMA model is adopted for this purpose, using Python as programming language. The results indicate that the K_t index has the potential to characterize GHI in locations with low climate variability.

However, in locations with climate variability, the performance need of additional atmospheric parameters to improve GHI.

This work can be used to analyze the climatic performance of photovoltaic. It also serves as a valuable tool to support operational and planning decisions in the solar energy sector in Algeria.

The future perspectives include extending the study to additional sites in Algeria to evaluate the model's generalizability and performing forecasts over longer time horizons. Additionally, hybrid models can be developed to improve forecasting accuracy in regions with high climatic variability.

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